Same grading scheme as Group C.

CS556-D Fall 2023 Midterm

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes.

Read carefully before answering! Write your answers on the white sheets given to you. Show all your work, not only the final answer.

Question 1 *(5 points)* If A and B are square and invertible, then AB is invertible, then $(AB)^{-1} = A^{-1}B^{-1}$.Explain.

False.
$(AB)^{-1} = B^{-1}A^{-1}$
And the latter is not equal to $A^{-1}B^{-1}$

Question 2 *(5 points)* If A is a 3-by-3 matrix and the equation $Ax = b$ has a unique solution, then A is invertible? Explain.

True. If system has a unique solution, then there is no free variables, so A is invertible.

Question 3 *(5 points)* If a system of linear equations has two different solutions, it must have infinitely many solutions. True or False? Explain.

True. Since a system can have 0, 1 or ∞ many solutions. If it has 2 solutions, then it has an infinite number of solutions.

Question 4 *(5 points)* If B, C, D are three matrices and $BC = BD$, then $C = D$. True or False?

False. $B(C-D) = 0$ for C different from D.

Question 5 *(5 points)* The columns of an invertible m-by-m matrix form a basis for $\mathcal{R}^m$. True or False? Explain.

True. The columns of an invertible matrix are linearly independent. A set of m independent vectors form a basis for $\mathbb{R}^m$.

Question 6 *(25 points)* Let $x_1 = \begin{bmatrix}2\\3\\4\end{bmatrix}$, $x_2 = \begin{bmatrix}1\\-1\\2\end{bmatrix}$, and $v = \begin{bmatrix}a\\b\\c\end{bmatrix}$.

a) Find a vector x_3 such that $\{x_1, x_2, x_3\}$ is a linearly independent set of vectors.

b) Find the determinant of the 3-by-3 matrix composed of columns x_1, x_2, x_3.

c) Find the rank of this matrix.

d) Find the inverse of this matrix.

e) Find a condition that a, b, c must satisfy for the set of the vectors $\{x_1, x_2, v\}$ to be linearly dependent.

a) In order for the set of vectors $\{x_1, x_2, x_3\}$ to be linearly independent the determinant of $A = [x_1\ x_2\ x_3]$ should be different from 0.

Let $x_3 = \begin{bmatrix}1\\-1\\0\end{bmatrix} \Rightarrow A = \begin{bmatrix}2 & 1 & 1\\3 & -1 & -1\\4 & 2 & 0\end{bmatrix}$

$$\det(A) = 1\cdot\begin{vmatrix}3 & -1\\4 & 2\end{vmatrix} + 1\begin{vmatrix}2 & 1\\4 & 2\end{vmatrix} = 10 - 0 = 10 \neq 0$$

b) $\det(A) = 10$ found on part a.

c) $\text{rank}(A) = 3$ since the vectors are linearly independent.

d) Using Gauss Jordan

$$A^{-1} = \begin{bmatrix}1/5 & 1/5 & 0\\-2/5 & -2/5 & 1/2\\1 & 0 & -1/2\end{bmatrix}$$

e) In order for the set to be linearly dependent $\Rightarrow \det(A) =$

$$\det\left(\begin{bmatrix}2 & 1 & a\\3 & -1 & b\\4 & 2 & c\end{bmatrix}\right) = a\begin{vmatrix}3 & -1\\4 & 2\end{vmatrix} - b\begin{vmatrix}2 & 1\\4 & 2\end{vmatrix} + c\begin{vmatrix}2 & 1\\3 & -1\end{vmatrix}$$

$$= 10a - 5c$$

$$\boxed{10a - 5c}$$

Question 7 *(25 points)* Given matrix $B = \begin{bmatrix} 1 & 1 & 4 & 5 \\ 2 & 1 & 5 & 5 \\ 0 & 1 & 3 & 2 \\ -1 & 1 & 2 & 2 \end{bmatrix}$

a) Is there a non-trivial solution of the equation $Bx = 0$?

b) Is there a vector b for which $Bx = b$ has no solutions? Write down such a vector if it exists.

c) Write down a vector c for which $Bx = c$ has at least one solution? Write down the complete solution of $Bx = c$.

- Identical to question 7 of Group C.

Question 8 *(25 points)* Consider the following system of linear equation for some constant k:

$$x + y - z = -2$$
$$3x - 5y + 13z = 15$$
$$x - 2y + 5z = k$$

a) For which value(s) of k does this system have one or infinitely many solutions?

b) Find all solutions for each value of k.

a) $\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 3 & -5 & 13 & 15 \\ 1 & -2 & 5 & k \end{array}\right] \xrightarrow[R3=R3-R1]{R2=R2-3R1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -8 & 16 & 21 \\ 0 & -3 & 6 & k+2 \end{array}\right] \xrightarrow[R3/-3]{R2/-8} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -2 & -21/8 \\ 0 & 1 & -2 & -(k+2)/3 \end{array}\right]$

$\xrightarrow{R_3=R_3-R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -2 & -21/8 \\ 0 & 0 & 0 & -\frac{(k+2)}{3} + \frac{21}{8} \end{array}\right]$

$-\frac{(k+2)}{3} + \frac{21}{8} = 0 \Rightarrow$ For infinitely many solutions

$k = \frac{47}{8}$

b) $\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 3 & -5 & 13 & 15 \\ 1 & -2 & 5 & 47/8 \end{array}\right] \xrightarrow[R3-R1]{R2-3R1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -8 & 16 & 21 \\ 0 & -3 & 6 & 63/8 \end{array}\right] \xrightarrow[R3/-3]{R2/-8} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -2 & -21/8 \\ 0 & 1 & -2 & -21/8 \end{array}\right]$

$\xrightarrow{R3-R2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -2 & -21/8 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R1-R2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 5/8 \\ 0 & 1 & -2 & -21/8 \\ 0 & 0 & 0 & 0 \\ x & y & z & \end{array}\right]$

$z \rightarrow$ free variable

- $x_{particular}$

$z = 0, y = -21/8, x = 5/8$

$x_p = \begin{bmatrix} 5/8 \\ -21/8 \\ 0 \end{bmatrix}$

- $x_{nullspace} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} s, \forall s \in \mathbb{R}$

$z = 1$

$y = 2$

$x = 1$

$X = X_p + X_n = \begin{bmatrix} 5/8 \\ -21/8 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} s, \forall s \in \mathbb{R}$

Question 9 *(5 points)* [Bonus] Please take a few minutes to share your thoughts on the course so far. Please be thoughtful and frank. Thank you.

- What do you like about the course so far?
- What suggestions do you have to improve this course?
- Do you have any other comments?